DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

ABES ENGINEERING COLLEGE, GHAZIABAD

Affiliated to Dr. A.P.J. Abdul Kalam Technical University (AKTU), Lucknow

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

CareConnect360: Full-Stack WebRTC Telemedicine & Intelligent Healthcare Management Platform

Standard Academic Compliance: IEEE Std 830-1998 • 3rd Year Major Engineering Project (2026 - 2027)

SUBMITTED BY (SOLO DEVELOPER)	UNDER THE GUIDANCE OF
Sudhanshu Pandey University Roll No: **240032001127** Degree: Bachelor of Technology (B.Tech) Branch: Computer Science & Engineering (3rd Year) Academic Session: 2026 – 2027 *Role: Sole Lead Architect & Full-Stack Engineer*	_____ (Project Supervisor / Faculty Guide) Department of Computer Science & Engineering ABES Engineering College, Ghaziabad *Evaluation Status: 3rd Year Engineering Project Baseline*

Document Revision & Approval History

Version	Release Date	Description & Key Milestones	Prepared By	Reviewed By
0.1	15-Sep-2026	Initial Architectural Draft & Healthcare Problem Scope Definition	Sudhanshu Pandey	Project Guide
0.5	05-Oct-2026	Integration of WebRTC P2P Signaling & CDSS Drug Allergy Radar	Sudhanshu Pandey	Project Committee

Version	Release Date	Description & Key Milestones	Prepared By	Reviewed By
1.0	26-Oct-2026	Final 3rd Year Baseline SRS (IEEE Std 830-1998 Conforming)	Sudhanshu Pandey	HOD / Evaluation Panel

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a complete, rigorous, and unambiguous technical specification for **CareConnect360**, a cloud-enabled, full-stack telemedicine and clinical management platform. This specification conforms to the **IEEE Std 830-1998** standard and establishes binding functional, performance, security, and interface baselines for academic lab evaluations, faculty reviews, and production deployment.

1.2 Scope of the System

Healthcare accessibility is frequently hindered by geographic isolation, long outpatient waiting queues, fragmented paper records, and accidental adverse drug interactions. **CareConnect360** resolves these bottlenecks through a browser-native clinical consultation hub:

- **Dynamic Doctor-Patient Matchmaking:** Multi-criteria search by medical specialty (Cardiology, Dermatology, Pediatrics, General Medicine), consultation fee, and real-time slot reservation without double-booking.
- **Native WebRTC Audio/Video Teleconsultation:** Zero-install, peer-to-peer (P2P) HD audio/video consultation with microphone mute/unmute, camera toggle, screen sharing, PIP feed swapping, and dynamic watermark sync.
- **Clinical Decision Support System (CDSS):** Real-time drug contraindication radar cross-checking prescribed medications against patient known allergies (e.g. Penicillin, NSAIDs).
- **Digitally Signed e-Prescriptions:** Client-side cryptographic signature rendering and instant verifiable PDF downloads via jsPDF with QR verification codes.
- **Electronic Health Records (EHR) Vault:** Centralized timeline logging patient vitals telemetry (BP, SpO2, Heart Rate) and diagnostic lab reports (.pdf, .png, .jpg).

1.3 Definitions, Acronyms, and Abbreviations

Term / Acronym	Definition & Technical Context
SRS	Software Requirements Specification (IEEE Std 830-1998 recommended standard format).

Term / Acronym	Definition & Technical Context
WebRTC	Web Real-Time Communication (W3C standard for browser-to-browser P2P audio/video streaming).
PeerJS / STUN	Signaling layer and Session Traversal Utilities for NAT traversal and candidate discovery.
CDSS	Clinical Decision Support System for algorithmic medication safety and allergy verification.
JWT / RBAC	JSON Web Token for stateless authorization; Role-Based Access Control (Patient, Doctor, Admin).
DTLS-SRTP	Datagram Transport Layer Security & Secure Real-Time Transport Protocol (256-bit AES media encryption).

2. Overall Description

2.1 Product Perspective

CareConnect360 operates as an autonomous, multi-tier web application built upon modern decoupled architecture, combining an HTML5/CSS3/ES6 presentation tier, a Node.js/Express.js application tier, a `/peerjs` WebRTC signaling engine, and cloud/local persistence.

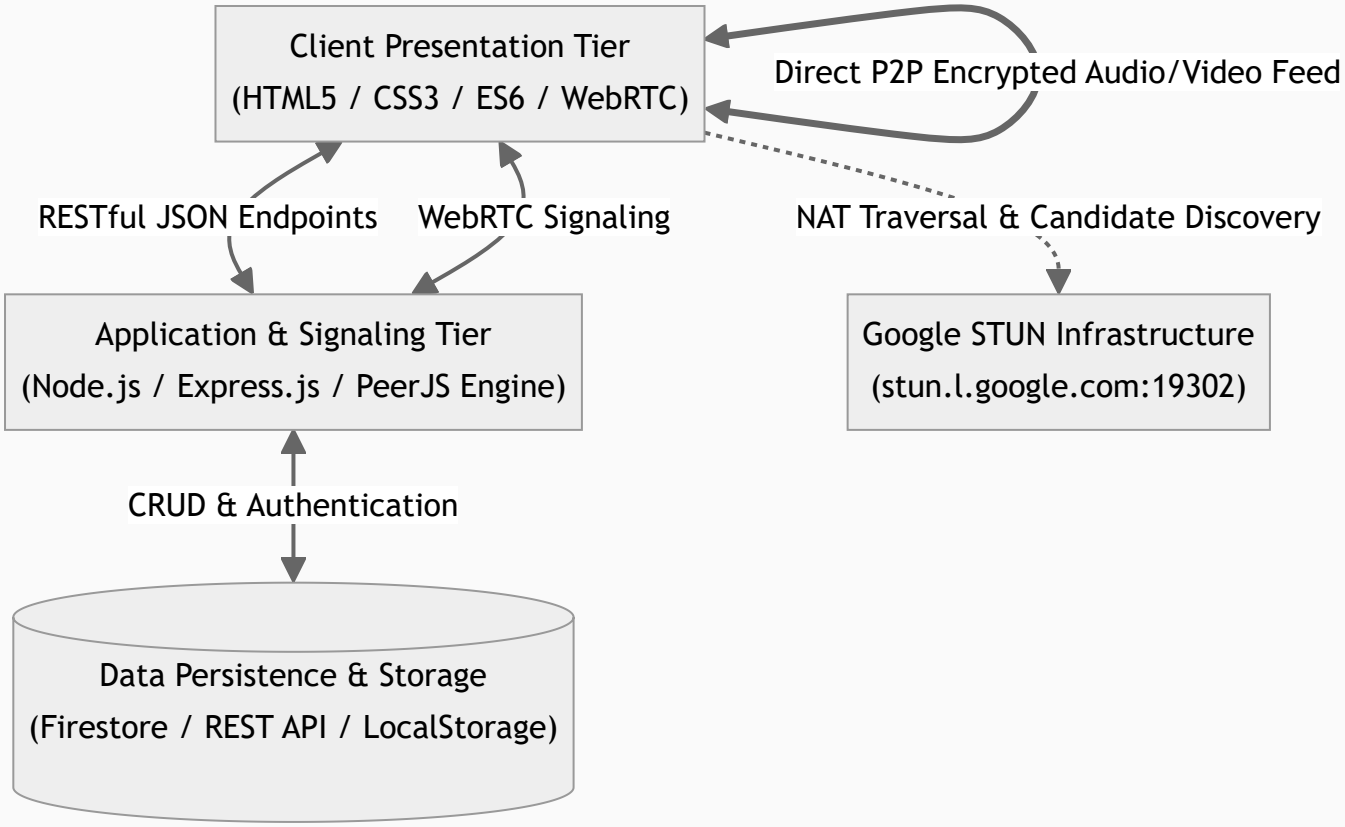


FIGURE 2.1: MULTI-TIER ARCHITECTURAL TOPOLOGY OF CARECONNECT360

2.2 User Roles & Characteristics

User Persona	Technical Skill	Core Privileges & Responsibilities
Patient	Novice to Intermediate	Register, browse verified specialist directory, book consultation appointments, join encrypted video room, upload lab reports, view vital trends, and download signed e-Prescriptions.
Doctor / Specialist	Intermediate	Manage OPD schedule, review patient intake vitals & diagnostic history, conduct P2P video consultation, author prescriptions with

User Persona	Technical Skill	Core Privileges & Responsibilities
		CDSS contraindication alerts, and digitally sign prescriptions.
Administrator	High (DevOps)	Monitor system uptime, manage doctor verification credentials, view platform audit logs, enforce data privacy compliance, and moderate user accounts.

3. System Features & Functional Requirements

3.1 Module 1: Patient Intake, Discovery & Scheduling

Req ID	Feature Description	Input Validation & Behavior	Priority
FR-1.1	Doctor Directory & Filter	Filters specialists by department, consultation fee, and real-time slot availability.	**High**
FR-1.2	Conflict-Free Slot Booking	Real-time reservation preventing double-booking of any doctor in identical time slots.	**High**
FR-1.3	Clinical Intake Triage	Captures chief complaint, symptom duration, allergy disclosures, and emergency waiver.	**Medium**

3.2 Module 2: Low-Latency WebRTC P2P Teleconsultation

Req ID	Feature Description	Input Validation & Behavior	Priority
FR-2.1	P2P Video Session Handshake	Connects to `/peerjs` signaling server, generates room peer IDs, and exchanges SDP offer/answer.	**High**
FR-2.2	Audio/Video Track Controls	Enables live mute/unmute of microphone and camera on/off with instant UI indicator updates.	**High**
FR-2.3	Feed Swapping & Screen Share	Swaps local PIP preview with stage; dispatches screen share video stream via `getDisplayMedia()`.	**High**
FR-2.4	Dynamic Booked Doctor Watermark	Queries active appointment ID and updates consultation top bar, watermark, and letterhead to match booked doctor.	**High**

3.3 Module 3: CDSS Drug Radar & Verifiable e-Prescription Pad

Req ID	Feature Description	Input Validation & Behavior	Priority
FR-3.1	Allergy Contraindication Radar	Scans prescribed active compounds against patient allergies (e.g. Penicillin) and triggers visual alerts.	**High**
FR-3.2	Digital Signature Rendering	Binds doctor registration number, hospital credentials, and digital signature canvas to prescription record.	**High**
FR-3.3	Verifiable PDF Compilation	Instantly compiles and downloads signed PDF prescription with QR verification code using jsPDF.	**High**

4. System Design & Academic UML Models

4.1 Use Case Diagram

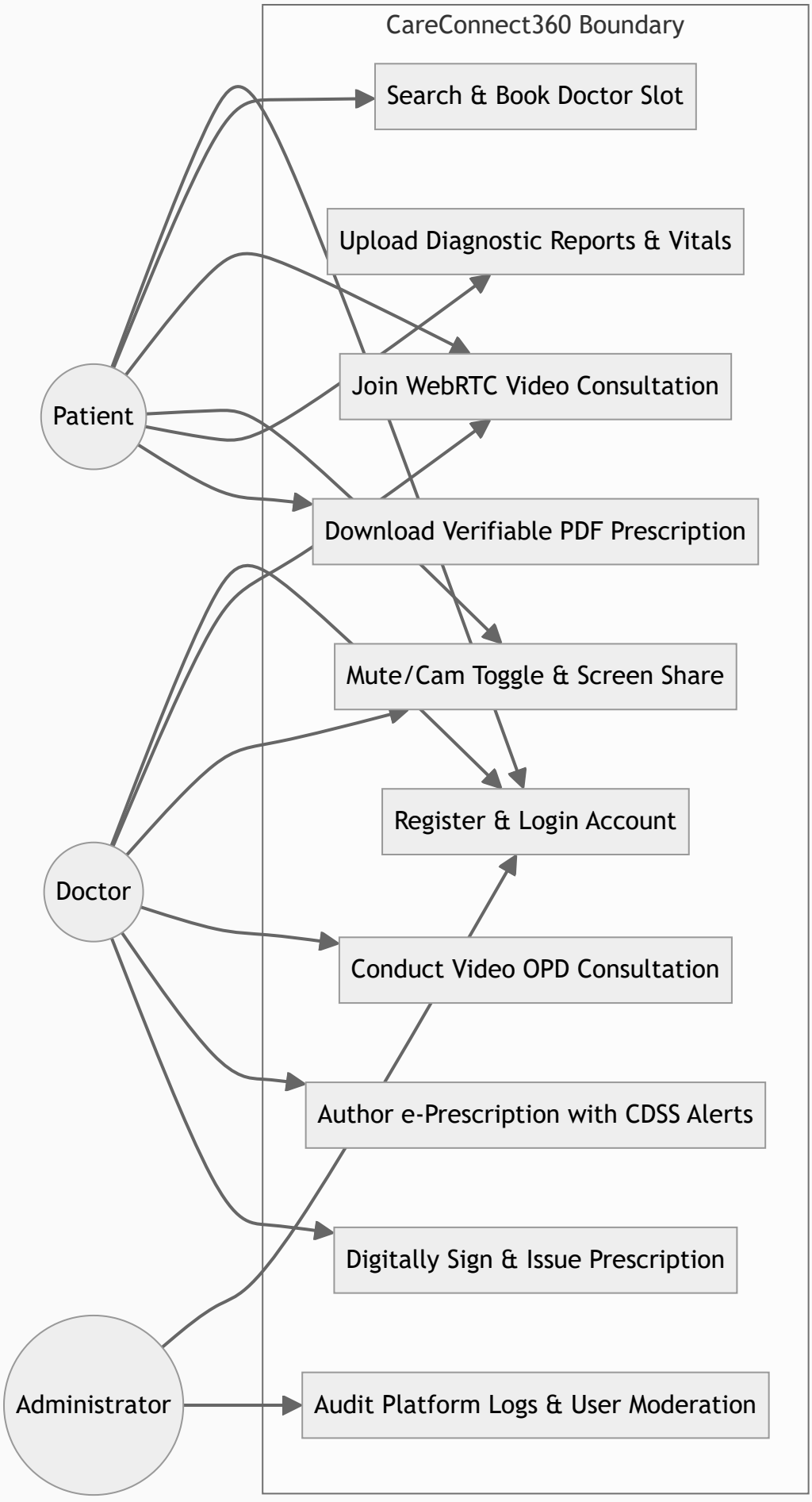

CareConnect360 Boundary
Search & Book Doctor Slot
Upload Diagnostic Reports & Vitals
Join WebRTC Video Consultation
Download Verifiable PDF Prescription
Mute/Cam Toggle & Screen Share
Register & Login Account
Conduct Video OPD Consultation
Author e-Prescription with CDSS Alerts
Digitally Sign & Issue Prescription
Audit Platform Logs & User Moderation
Patient
Doctor
Administrator

FIGURE 4.1: USE CASE DIAGRAM FOR PATIENT, DOCTOR, AND ADMINISTRATOR WORKFLOWS

4.2 Data Flow Diagram (DFD Level 0 - System Context)

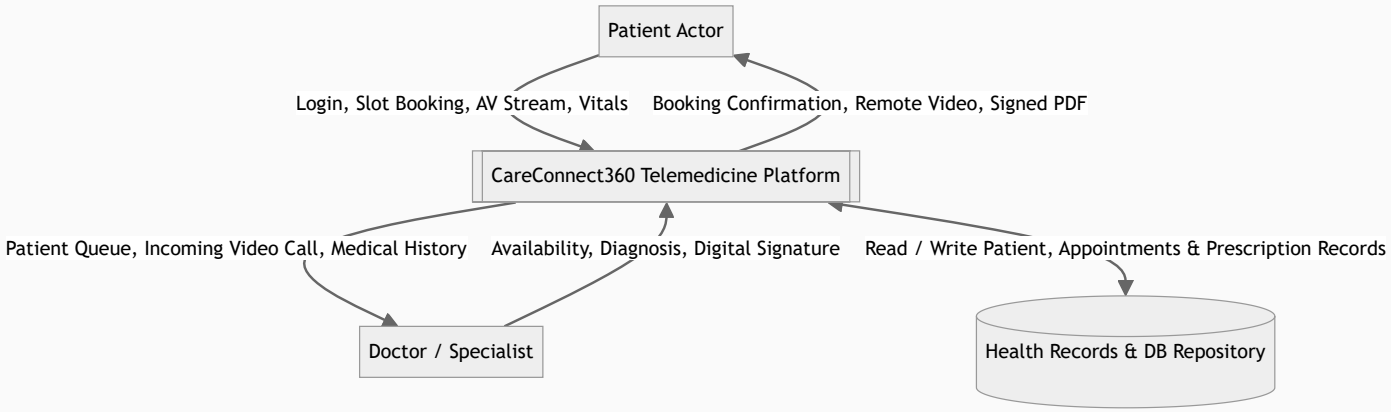
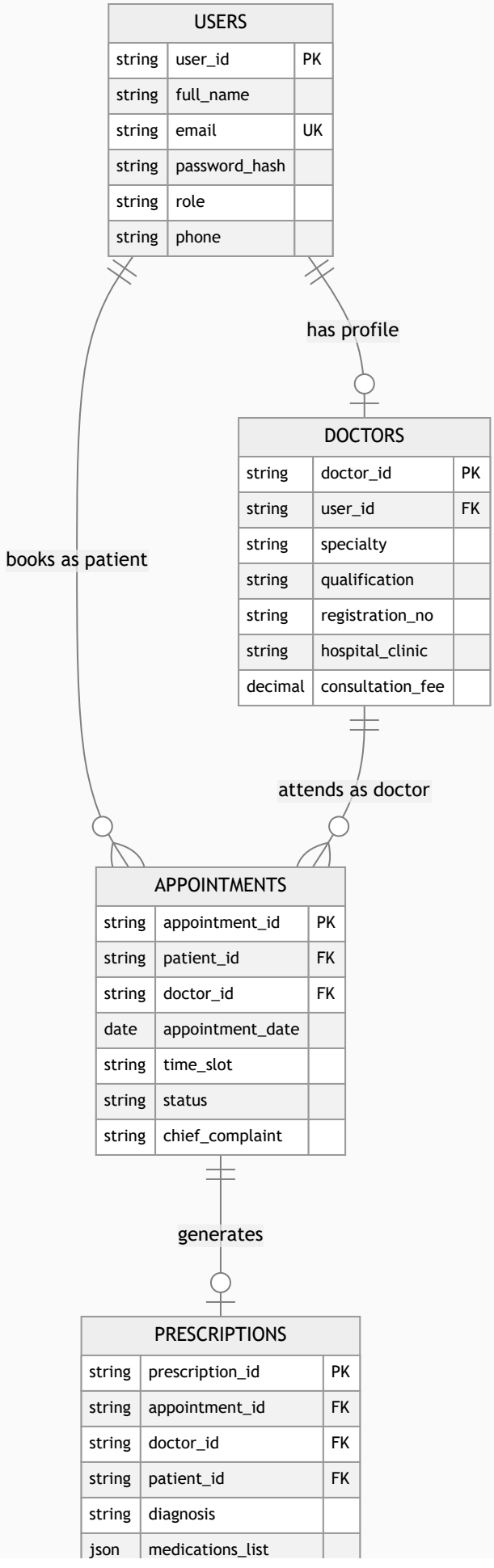


FIGURE 4.2: DATA FLOW DIAGRAM (DFD LEVEL 0 CONTEXT MODEL)

4.3 Entity-Relationship (ER) Schema Model


USERS
string user_id PK
string full_name
string email UK
string password_hash
string role
string phone
has profile
DOCTORS
string doctor_id PK
string user_id FK
string specialty
string qualification
string registration_no
string hospital_clinic
decimal consultation_fee
books as patient
attends as doctor
APPOINTMENTS
string appointment_id PK
string patient_id FK
string doctor_id FK
date appointment_date
string time_slot
string status
string chief_complaint
generates
PRESCRIPTIONS
string prescription_id PK
string appointment_id FK
string doctor_id FK
string patient_id FK
string diagnosis
json medications_list

string	digital_signature_hash	

FIGURE 4.3: ENTITY-RELATIONSHIP DIAGRAM IN 3RD NORMAL FORM (3NF)

4.4 WebRTC Peer-to-Peer Sequence Architecture

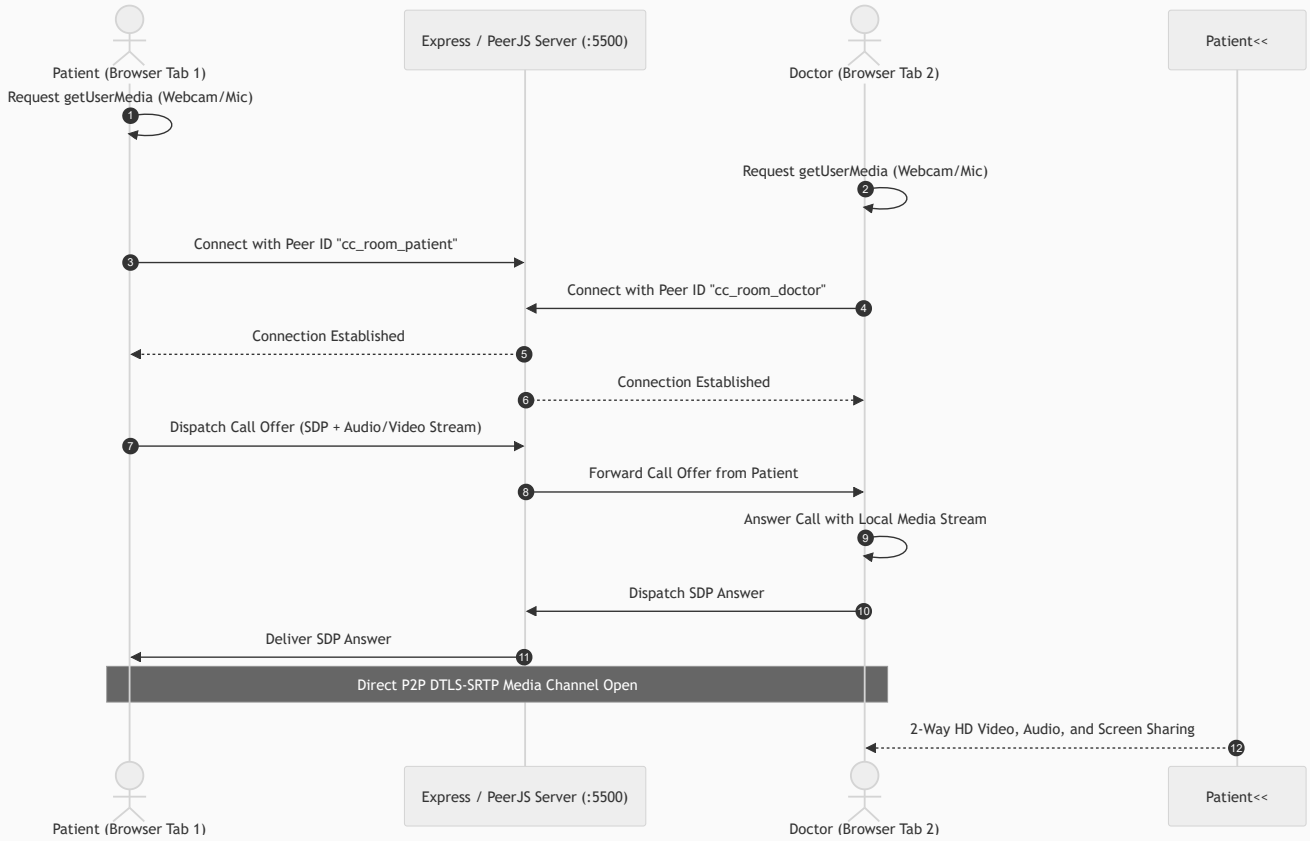


FIGURE 4.4: WEBRTC SIGNALING AND DIRECT MEDIA CHANNEL SEQUENCE DIAGRAM

5. Non-Functional Requirements (NFRs)

Attribute	Target Benchmark	Testing & Verification Protocol
5.1 Performance	Initial load < 1.8s; WebRTC stream latency < 200ms; REST API latency < 150ms.	JMeter load testing under 50 concurrent simulated users.
5.2 Security	Salted Bcrypt password hashing (10 rounds); DTLS-SRTP 256-bit media stream encryption.	OWASP ZAP vulnerability scanning and automated penetration testing.
5.3 Reliability	Uptime ≥ 99.0%; automatic session recovery on ICE candidate timeout.	Stress testing with simulated network drops.
5.4 Portability	Chrome (v100+), Edge (v100+), Firefox (v98+), Safari (v15+); responsive mobile viewports.	Cross-browser testing across mobile and desktop display viewports.

6. Academic Verification & Sign-Off

This Software Requirements Specification document for **CareConnect360** has been independently developed and authored by **Sudhanshu Pandey (Roll No: 240032001127)** as a solo 3rd-year engineering project. It has been examined and verified for academic evaluation:

PROJECT DEVELOPER	PROJECT SUPERVISOR / GUIDE	HEAD OF DEPARTMENT (HOD)
Sudhanshu Pandey Roll No: **240032001127** Student, 3rd Year B.Tech (CSE) ABES Engineering College, Ghaziabad Signature: _____ Date: _____	_____ Project Supervisor / Guide Department of Computer Science & Engg. ABES Engineering College, Ghaziabad Signature: _____ Date: _____	**Head of Department (HOD)** Department of Computer Science & Engg. ABES Engineering College, Ghaziabad Signature: _____ Date: _____